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PALEOMAGNETIC RESULTS FROM EUROPE[1]

A. E. M. NAIRN[2]

ABSTRACT

The results of investigations on a variety of late Paleozoic and Mesozoic rocks from western Europe are described and are incorporated with results already published predominantly from Great Britain. Mean pole positions have been obtained from the Triassic, Permian, and Devonian periods, for which numerous results are available. The angular distance between the mean Triassic pole and a similar American pole lends force to the separation of the two continents, while the greater separation of Permian poles implies that separation commenced in earlier times. Because of the problems, still unresolved, of the European Carboniferous, this drift history cannot as yet be checked in earlier times.

I. INTRODUCTION

In 1954 suggested positions of the pole for various geological epochs, based on paleomagnetic measurements, were linked together to form a "polar-wandering curve" (Creer *et al.*, 1954). A more detailed curve, published by the same authors (1957), incorporated previously unpublished work of Belshé (1957) and some results of Clegg *et al.* (1954) and figured a set of pole positions obtained from American rocks.

There were, however, geological periods still unrepresented, and the areas from which samples had been obtained were comparatively restricted. The results now to be described are an attempt to expand and augment these original results. The collections were made between 1955 and 1957 from rocks ranging in age from Devonian to Tertiary and obtained principally from Britain, France, and Germany. Where possible, red-bed lithologies and extrusive igneous rocks were collected, but as these do not occur in all the formations sampled, drab arenaceous lithologies were also collected.

The most common interpretation of paleomagnetic results is based on the assumptions that, over a period of greater than a few thousand years, the earth's magnetic field approximates to a geocentric axial dipole (that is, secular variation is averaged out), and this has always been the case. It is not proposed to consider the validity of these assumptions or the origin or methods of testing the stability of remanent magnetization, as these have been fully dealt with elsewhere (Graham, 1949; Runcorn, 1955, 1956*a;* Blackett, 1956; Irving, 1956; Creer, 1957; Irving and Runcorn, 1958; Collinson and Nairn, 1959). It follows from these assumptions that, where the pole position for a group of rocks is significantly different from the theoretical dipole field, then there has been relative movement of the crust with respect to the rotation axis, whether movement is of the polar-wandering type (complete shell) or of the continental-drift (individual fragments) type. Consequently it is necessary to show that the pole positions given are valid by collecting from as wide an area and as many lithologies as possible. It automatically follows that, if pole polisitions for different geological periods can be established, paleomagnetism can function as a dating technique (Armstrong, 1957; Blundell, 1957; Blundell and Read, 1958).

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II. EUROPEAN PALEOMAGNETIC RESULTS

1. TERTIARY

Very many Tertiary rocks have been examined from many areas (table 1), the vast majority of the rocks being igneous—either lavas or intrusives. Their importance derives from the fact that whether possessing normal, that is, north pole pointing down in the Northern Hemisphere, or reversed

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TABLE 1

SUMMARY OF EUROPEAN TERTIARY PALEOMAGNETIC DATA

Site			Age and Lithology	Mean Declination (Degrees)	Mean Inclination (Degrees)	α*	K†	N‡	S§	Ancient Pole		$d\psi$‖	$d\chi$#	Reference
Iceland:														
S.W. and N. Iceland	65.2 N.	20 W.....	Miocene lavas	1.5	+77.8	5.5	8.4	102	..	89 N.	5 E.	10	10	Hospers, 1954*a*
N. Iceland	65.2 N.	20 W.....	Miocene lavas	180.5	−70	....	....	...	..	.....	.....	..	..	Hospers, 1954*a*
Britain:														
Mull	56.5 N.	6 W.....	Eocene ring dikes	25	+61	....	....	...	..	.....	.....	..	..	Vincenz, 1954
			Eocene lavas	166	−73	....	....	...	..	.....	.....	..	..	Bruckshaw and Vincenz, 1954
N. England	55.5 N.	2 W.....	? Miocene dikes	175	−65	....	....	...	..	.....	.....	..	..	Bruckshaw and Robertson in Vincenz, 1954
N. Ireland	55 N.	6 W.....	Eocene lavas	14	+60	5	....	...	..	74 N.	133 E.	6	8	Hospers and Charlesworth, 1954
Germany:														
Bonn	50.5 N.	7 E.....	Oligocene-Miocene intrusions	188	−63	14	16.3	48	18	.....	.....	17	22	
Vogelsberg	50.5 N.	9 E.....	Intrusions and lavas	177	−57	12.3	10	43	16	.....	.....	12	17	
			Intrusions and lavas	9	+73	11	22.8	29	9	.....	... ..	7	12	
France:														
Auvergne	45 N.	3 E.....	Plio-Pleistocene dikes, necks, lavas	165	−61	....	....	...	..	.....	.....	..	..	Roche, 1951
			Mio-Pliocene dikes, necks, lavas	187	−50	....	....	...	..	.....	.....	..	..	Roche, 1951
			Miocene dikes, sills, necks	200	−53	....	....	...	..	.....	.....	..	..	Roche, 1950
			Oligocene lavas	180	−73	....	....	...	..	.....	.....	..	..	Roche, 1950
				21	+57	....	....	...	..	.....	.....	..	..	Roche, 1950

Not significantly different from present pole

* α = radius of circle of confidence in degrees (at 95 per cent level).

† K = measure of the scatter of results (small κ, high scatter).

‡ N = number of discs or cylinders measured.

§ S = number of disks or cylinders measured.

‖ $d\psi$ = semi-minor axis of ellipse of confidence about the mean pole position.

$d\chi$ = semi-major axis of ellipse of confidence about the mean pole position.

TABLE 2

SUMMARY OF EUROPEAN CRETACEOUS AND JURASSIC PALEOMAGNETIC DATA

Site			Age and Lithology	Mean Declination (Degrees)	Mean Inclination (Degrees)	α	K	N	S	Ancient Pole		$d\psi$	$d\chi$	Reference
Britain:														
Sussex	54 N.	0.5 E.....	Lower Cretaceous "Iron Grit"	2.6	+68	5.7	20	35	21	86 N.	26 W.	8	96	
				184.5	−72	6.5	10.3	17		84 N.	182 E.	10	11	
Isle of Wight	54 N.	0.5 E.....	Lower Cretaceous Wealden sandstones and clays	2	+59	...	.....	97	19	75 N.	9 W.	....	....	Wilson, 1959
E. and W. Scotland	54 N.	0.5 E.....	Middle Jurassic limestones and sandstone	234.4	−66	6.6	33.5	14	4	57 N.*	76 E.	8.8	10.8	Nairn, 1957*a*
Germany and Austria:														
Austro-German frontier	47.5 N.	12.5 E.....	Middle Jurassic radiolarite	36.7	47.8	5.5	100	21	15	58.5 N.	128 E.	4.7	7.2	Hargraves and Fischer, 1959
			Lower Jurassic limestone	47.9	50.6	6.5	70.7	30	16	53 N.	112 E.	5.9	8.8	Hargraves and Fischer, 1959

* Recalculated pole position differs from previously published figure.

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polarity, the pole positions derived from the mean directions of magnetization for all results from Miocene and younger rocks do not appear to differ significantly (Fisher, 1953) from the geographical poles. This is a strong argument in favor of the dipole assumption. Hospers (1955) suggested that in Eocene times the divergence of the mean direction from the dipole direction, although small, was significant (further results in Wilson, 1959).

The geographical distribution of the sites and the mean directions of magnetization at these sites are clearly illustrated in figure 1. The only new results are those from the Bonn area (fig. 2)[3] (see also Nairn and Kürsten, 1959) and those from the Vogelsberg (fig. 3) (see also Angenheister, 1956)

[3] Stereographic projections are used throughout, with the following conventions: solid circle denotes a north-seeking pole pointing down; an open circle when pointing up. Circles in asterisks are mean directions, while the theoretical dipole field at the site where the collection was made is shown by a plus sign.

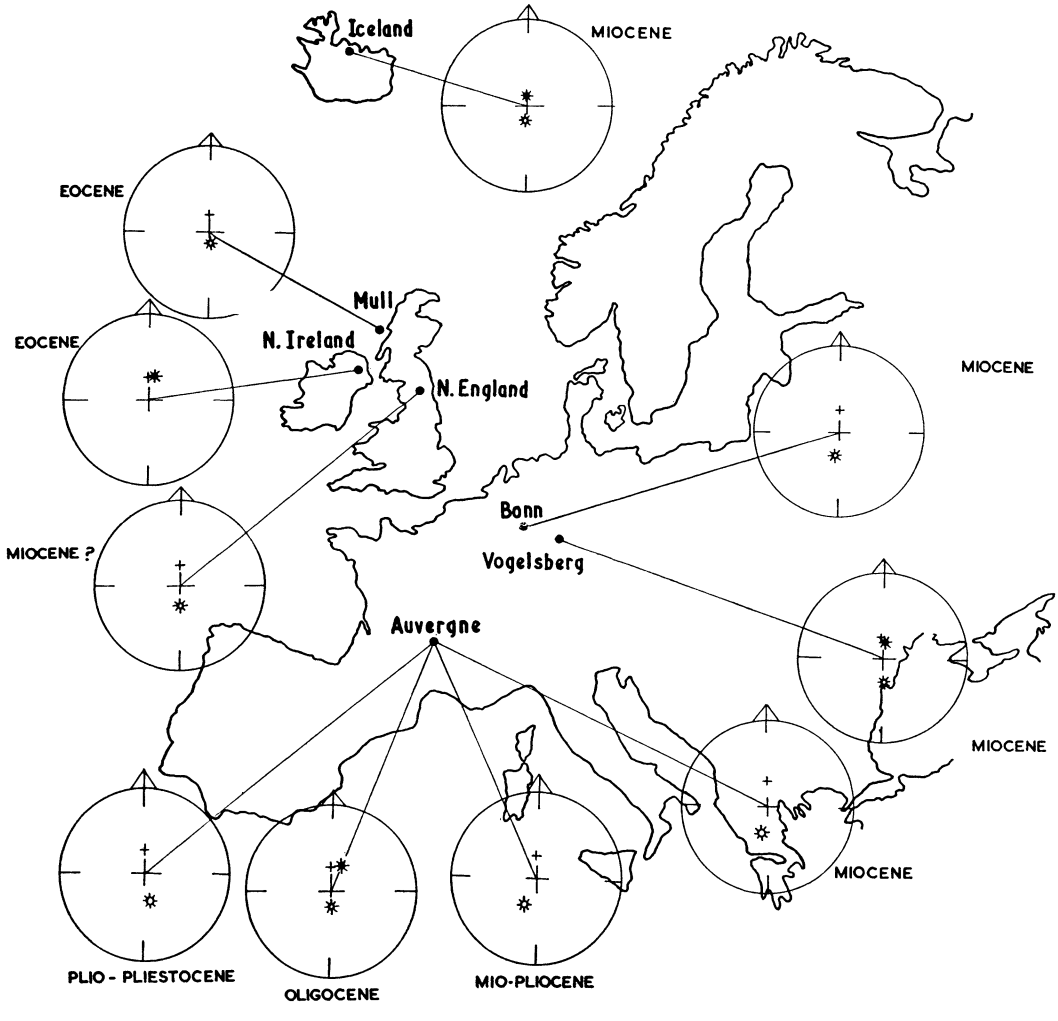


FIG. 1.—Summary of all paleomagnetic studies of Tertiary rocks showing areas from which collections were obtained and the mean magnetic directions for each group of a site. The circle of confidence when known is drawn only where it exceeds 10°.

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illustrating the scatter of sample mean directions (up to four measurements per sample) from which the mean is calculated.

So far as is known within the Tertiary, the numbers of reversely and normally magnetized rocks are about equal, but it is not yet possible to state on how many occasions the earth's magnetic field reversed its polarity. Hospers (1954*b*) suggested that such field reversals might be used for correlation purposes, and, in Iceland, mapping of normal and reversed groups of lavas has been carried out (Einarsson, 1957; Sigurgeirsson, 1957).

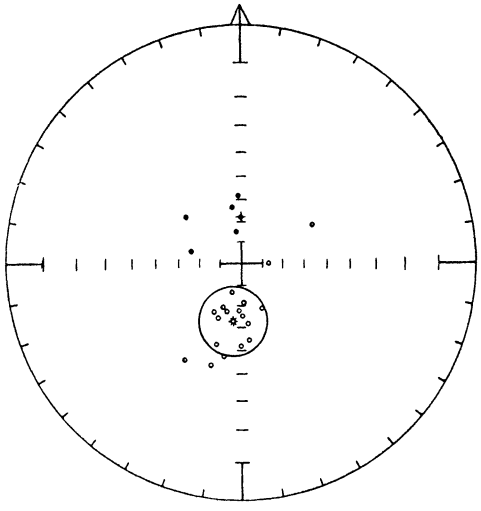


FIG. 2.—The directions of magnetization, mean, and circle of confidence of Tertiary plugs from the region around Bonn, Germany; sample means plotted.

2. CRETACEOUS

Over Europe during the Cretaceous and also during Jurassic times, the rocks deposited are lithologically unsuited for paleomagnetic work, with the result that, despite detailed sampling, few satisfactory measurements have been made. Thus in the Wealden area of southern England the lower Cretaceous with its variety of lithologies yielded but a single horizon with a stable remanent magnetization. This horizon, the "Iron Grit" (Kirkcaldy, 1935), is only a few inches thick, yet it showed both normal and reversed directions (fig. 4). It derived its magnetization as the result of the formation of the ferromagnetic phase by chemical change. The pole positions corresponding to the mean directions of the normally and reversely magnetized groups do not differ significantly from the present geographic

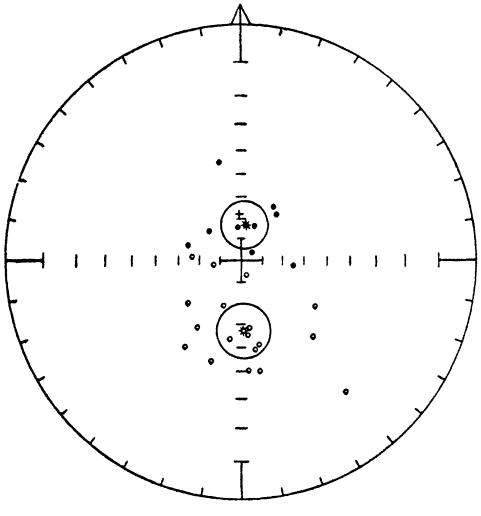


FIG. 3.—The directions of magnetization, mean, and circle of confidence of Tertiary lavas and intrusives from the Vogelsberg, Germany; sample means plotted.

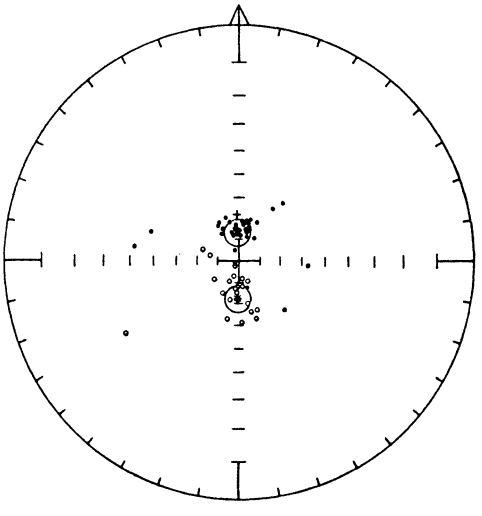


FIG. 4.—The directions of magnetization of the Cretaceous "Iron Grit" of Sussex; all measurements plotted.

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poles. However, the pole positions corresponding to the direction of magnetization found by Wilson (1959) in rocks from the Isle of Wight are further away (table 2), and the directions of magnetization were, he concluded, midway between those of Triassic and Eocene rocks.

3. JURASSIC

Arenaceous lithologies sampled in Luxemburg, northern France, as well as in Yorkshire, Skye, and Sutherland, England, have proved to be predominantly unstable magnetically (Nairn, 1957*a*). The single, tentative result obtained mainly from a few limestone samples is given in table 2. Hargraves and Fischer (1959) have since obtained better results from red limestone and radiolarite samples in the Alps.

4. TRIASSIC

The previously published Triassic results (Clegg *et al.*, 1954; Creer *et al.*, 1954) were from rocks of Keuper age. In Europe, extensive collecting from the lower Bunter beds of France and Germany was carried out. In general, it was found that the lower Buntsandstein was normally magnetized (figs. 5, 6, and 7) with a reversal apparently occurring somewhere about the boundary between middle and upper Buntsandstein (fig. 8). Although all the Grès vosgien samples collected were also normally magnetized (fig. 9), Clegg *et al.* (1957) have found reversely magnetized rocks within the same formation.

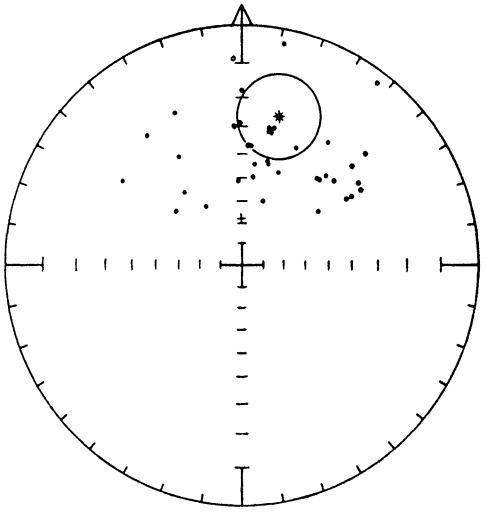


FIG. 6.—The directions of magnetization, mean, and circle of confidence of the middle Buntsandstein from northeast of Frankfurt am Main, Germany; all measurements plotted.

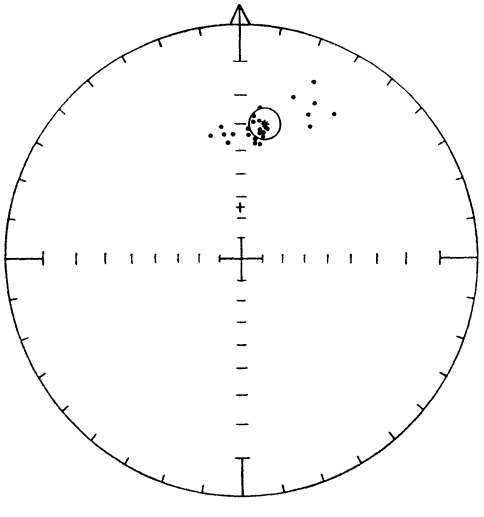


FIG. 5.—The directions of magnetization, mean, and circle of confidence from the Buntsandstein near Lörrach, Germany; all measurements plotted.

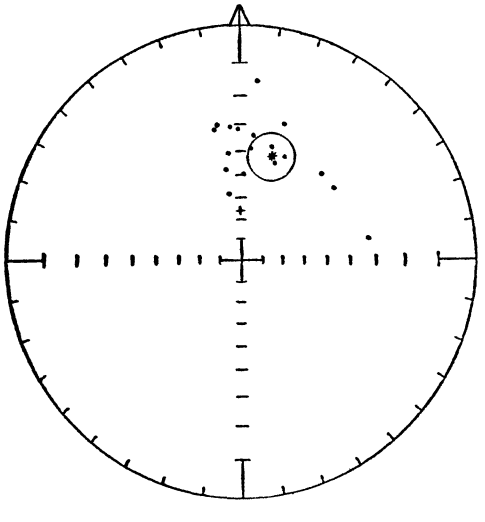


FIG. 7.—The directions of magnetization, mean, and circle of confidence of the middle Buntsandstein, Freudenstadt, Germany; all measurements plotted.

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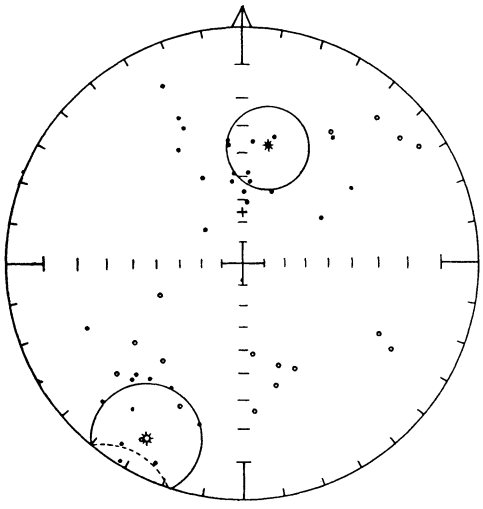


FIG. 8.—The directions of magnetization, mean, and circles of confidence of the middle Buntsandstein (normally magnetized) and the middle-upper Buntsandstein (reversely magnetized) from the Saar–Pfalz–Baden-Baden areas of Germany; all measurements plotted.

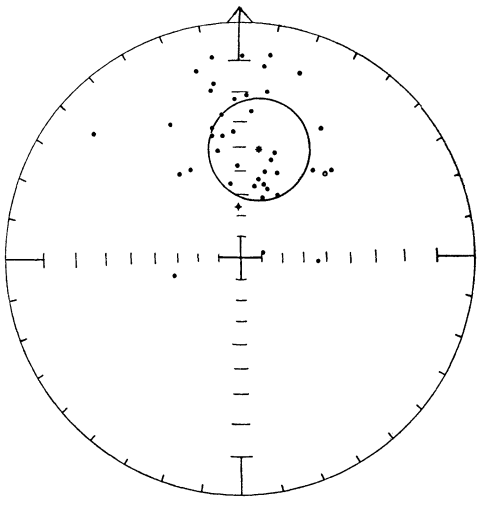


FIG. 9.—The directions of magnetization, mean, and circle of confidence of the Triassic Grès vosgien in eastern France; all measurements plotted.

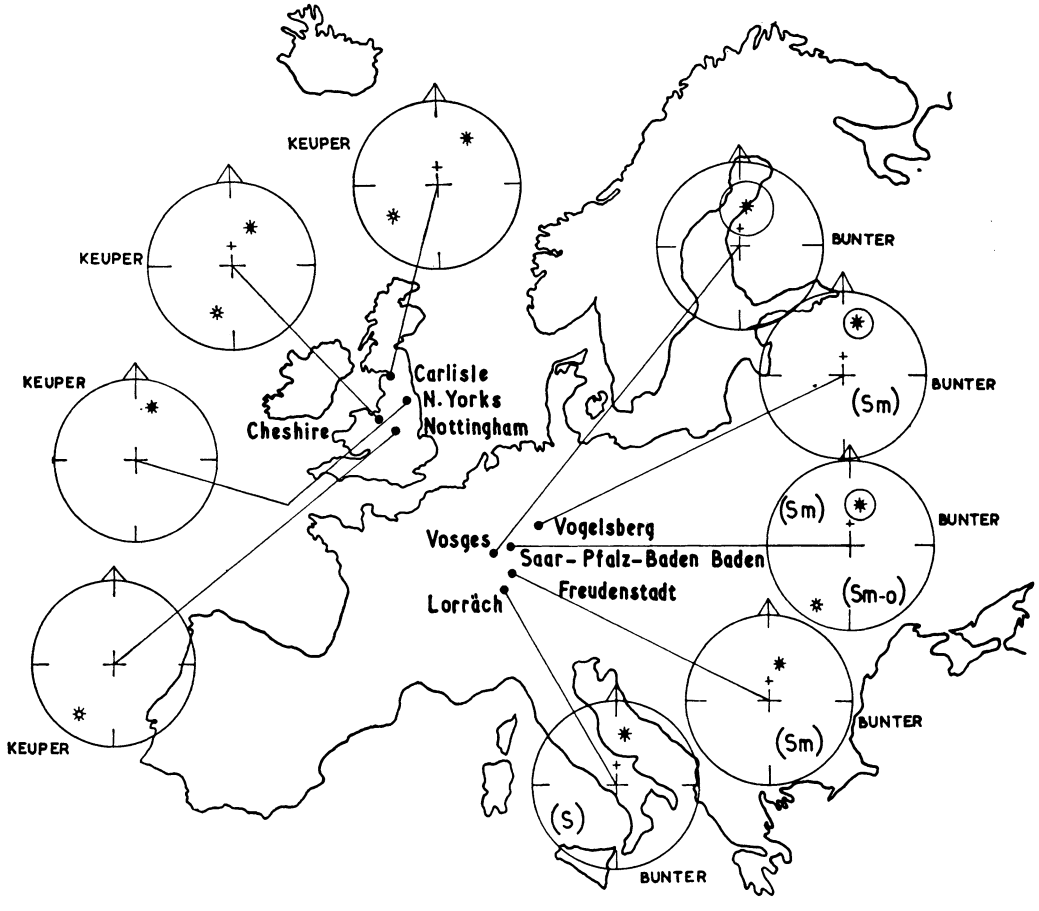


FIG. 10.—Summary of all known Triassic paleomagnetic results, indicating the mean magnetic directions for each site. The circle of confidence when known is drawn only where it exceeds 10°. *So* = upper Buntsandstein; *Sm* = middle Buntsandstein.

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TABLE 3

SUMMARY OF EUROPEAN TRIASSIC PALEOMAGNETIC DATA

	Site			Age and Lithology	Mean Declination (Degrees)	Mean Inclination (Degrees)	α	K	N	S	Ancient Pole	dψ	dχ	Reference
Britain:														
C_1	Carlisle Basin	55 N.	3 W.....	Keuper marl Series	34	+28	3.0*	....	37	3	42 N. 130 E.	2	3	Clegg *et al.*, 1954
C_2	Carlisle Basin	55 N.	3 W.....	Keuper marl Series	236	−27	1.5*	....	37	2	30 N. 129 E.	1	2	Clegg *et al.*, 1954
Ch_1	Cheshire	53 N.	3 W.....	Keuper marl Series	30	+39	.....	....	305	22	52 N. 129 E.	1	2	Clegg *et al.*, 1954
Ch_2	Cheshire	53 N.	3 W.....	Keuper marl Series	198	−28	1.5*	....	37	2	49 N. 150 E.	1	2	Clegg *et al.*, 1954
N	Nottingham, R. Trent	53 N.	1 W.....	Keuper marl Series	215	−16	1.5*	....	119	8	37 N. 134 E.	1	2	Clegg *et al.*, 1954
Y	N. Yorks., R. Leven	54 N.	1 W.....	Keuper marl Series	16	+22	1.5*	....	9	2	46 N. 156 E.	1	2	Clegg *et al.*, 1954
Germany:														
P_1	Pfalz	45.5 N.	8 E.....	Buntsandstein (S_m)	12.5	+37	14.5	4.8	25	5	63 N. 161 E.	10	17	
P_2	Pfalz	45.5 N.	8 E.....	Buntsandstein (S_m−o)	209	−10	18.2	3.4	26	6	42 N. 147 E.	9	18	
V_1	Vogelsberg	50 N.	9 E.....	Buntsandstein (S_m)	14	+25	12	4.5	40	8	51 N. 167 E.	7	13	
B	Baden-Baden-Freiburg	49 N.	8 E.....	Buntsandstein (S_m)	17	+40	9.7	13.7	18	4	61 N. 155 E.	7	12	
L	Lörrach	47.5 N.	7.5 E.....	Buntsandstein (S)	10	+30	4.7	4.7	24	4	58 N. 169 E.	3	5	
France:														
V	Vosges	49 N.	7 E.....	Grès vosgien	10	+40	23.2	9.2	46	9	62 N. 167 E.	17	28	
											Mean pole position = 49 N. 147 E.			

* Radius of circle of confidence at 50 per cent level.

TABLE 4

SUMMARY OF EUROPEAN PERMIAN PALEOMAGNETIC DATA

	Site			Age and Lithology	Mean Declination (Degrees)	Mean Inclination (Degrees)	α	K	N	S	Ancient Pole	dψ	dχ	Reference
Norway:														
N	Oslo	60 N.	10.5 E.....	Permian lavas and rhombo-porphyry	202	−33	....	....	..	12	46 N. 165 E.	..	..	Rutten *et al.*, 1957
Britain:														
A_1	Ayrshire	55.5 N.	4.5 W	Mauchline lavas	180	− 4	8	10.3	34	..	36 N. 175 E.	4	8	Du Bois, 1957
A_2	Ayrshire	55.5 N.	4.5 W.....	Mauchline sandstone	187	− 6	11.6	7.0	26	..	37 N. 166 E.	6	12	Du Bois, 1957
A_3	Ayrshire	55 5 N.	4.5 W.....	Minor (kylitic) intrusions	181	+ 7	10	24.4	..	11	32 N. 174 E.	5	10	Armstrong, 1957
D	Exeter	51 N.	4 W.....	Exeter volcanic series	189	− 9	20	15	33	5	43 N. 164 E.	10	20	Creer, 1957
Germany:														
S	Saar	49 N.	7 E.....	Rotliegendes (R) sandstones, lavas	183	− 5	5.7	16.6	40	9	43 N. 183 E.	3	6	
P	Pfalz	49 N.	7.5 E.....	Rotliegendes (R_o) sandstones, lavas	186	+ 4	5.5	13.3	53	13	38 N. 179 E.	3	6	
B	Baden-Baden	49 N.	8 E.....	Rotliegendes (R_o, R_m) sandstones, lavas	182	− 7	5.6	23.4	33	10	45 N. 185 E.	3	6	
F	Freiburg	48 N.	8 E.....	Rotliegendes (R) sandstones	157	+10	26.7	2.6	18	4	33 N. 150 W.*	14	27	
France:														
V	Vosges	48 N.	6 E.....	Saxonian "rhyolites"	193	− 7	4 5	22	49	14	43 N. 168 E.	2	5	Nairn, 1957
M	Morvan	47 N.	4 E.....	Saxonian sandstone	197	+ 6	4.1	93	14	3	38 N. 162 E.	2	4	Nairn, 1957
E_1	Esterel	43 5 N.	7 E.....	Permian pyromerides	210	−16	....	....	..	5	46 N. 142 E.	..	..	Roche, 1957
E_2	Esterel	43.5 N.	7 E.....	Permian dolerites	175	−13	....	....	..	3	54 N. 165 W.	..	..	Roche, 1957
E_3	Esterel	43.5 N.	7 E.....	Permian rhyolite (R3)	217	−23	....	....	..	14	45 N. 131 E.*	..	..	Rutten *et al.*
											Mean pole position = 42 N. 169 E.			

* Not included in calculation of mean pole position.

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All the available results are summarized in table 3, while in figure 10 the mean directions of magnetization and distribution of the sites are illustrated.

Some of the French and German rocks show rather scattered directions of magnetization, while in general the intensity of magnetization is low. As the field tests of stability cannot be made, the beds in general having a very low dip, it is difficult to be certain of the stability of some samples. However, the recurrence of reversals suggests that some at least are stable, and there is no doubt of the stability of the Keuper samples of Clegg *et al.* (1954). All samples whose directions of magnetization lay closer to the dipole direction when the rock was restored to its field position than when measured with respect to the bedding plane were rejected (see Collinson and Nairn, 1959, fig. 6).

5. PERMIAN

This period is one most suited to paleomagnetic investigation with a profusion of red beds and extrusives and intrusives of varied lithological types; consequently it

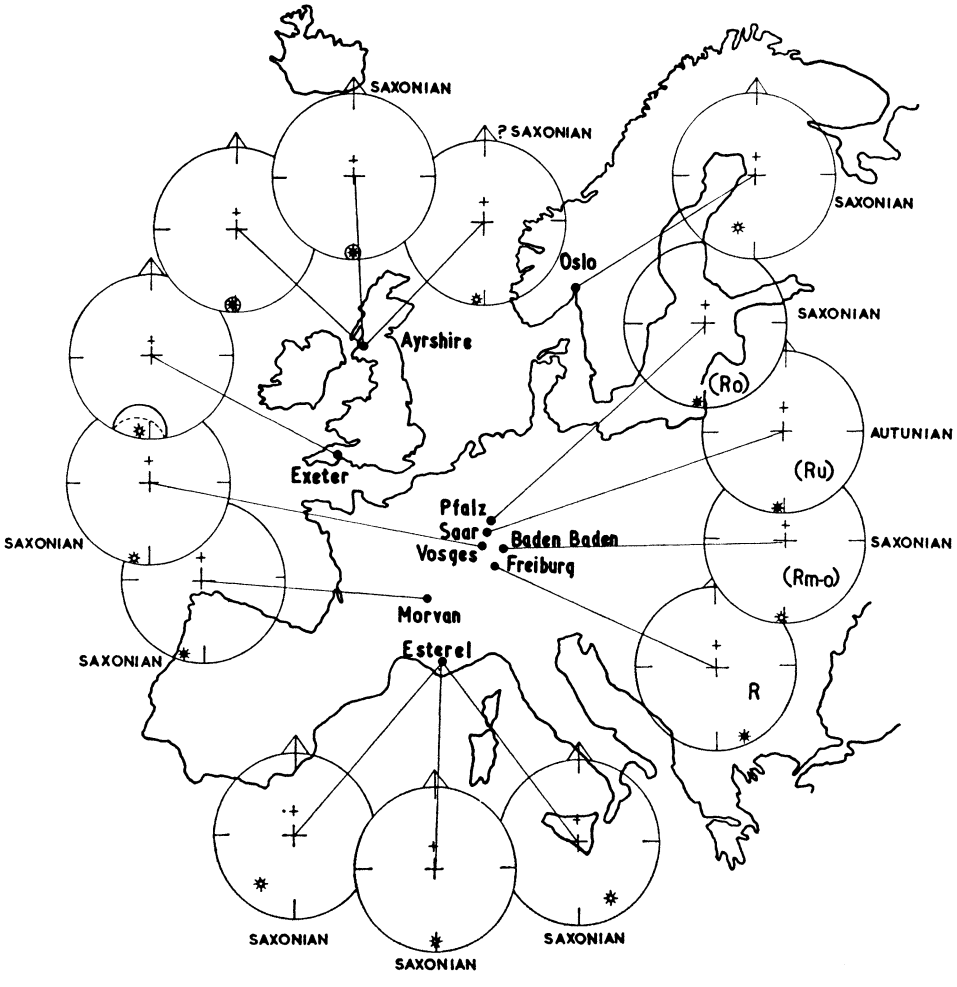


FIG. 11.—Summary of all known Permian paleomagnetic results, indicating the mean magnetic directions for each group of sites. The circle of confidence when known is drawn only where it exceeds 10°. *Ro* = upper Rotliegendes; *Rm* = middle Rotliegendes; *Ru* = lower Rotliegendes.

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is not surprising that so many results have have been obtained (see fig. 11). Three new results from the Saar, Pfalz, and Baden-Baden regions of Germany (figs. 12, 13, and 14) represent collections of a variety of lavas, sandstones, and marls. The pattern of a southerly declination and low inclination (table 4) is as surely convincing as the Tertiary one of high inclinations, although no reversals have been reported in Permian rocks. It must be noted, however, that these results are in general from Lower Permian rocks only, for the Thuringian is not represented.

6. CARBONIFEROUS

In contast to Permian, there is a considerable scatter in the Carboniferous results. The initial results of Belshé were from the Upper Carboniferous Coal Measures (quoted in Creer *et al.*, 1957; Belshé, 1957) and from lavas in the D_2 zone of the Lower Carboniferous. The directions of magnetization of the Staffordshire Keele beds (fig. 15) of top Westphalian or even low Stephanian age give a pole position similar to Belshé's, as does that obtained from samples from the Southdean basanite plug (fig. 16), which outcrops near Jedburgh. These contrast strongly with the younger Stephanian pole positions obtained from sediments collected in the Saar (fig. 17) and in the Blanzy (fig. 18) and St. Étienne (fig. 19) basins in France, which not surprisingly

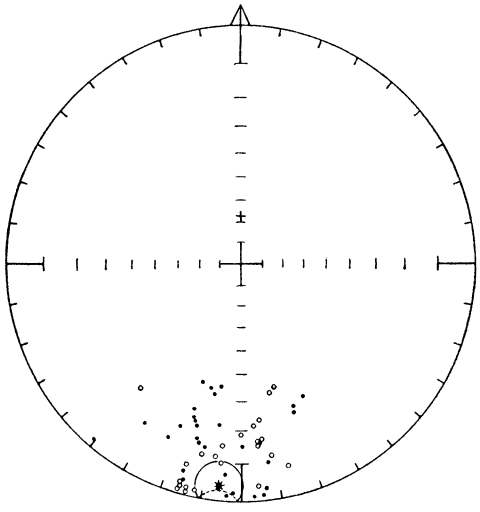


FIG. 13.—The directions of magnetization, mean, and circle of confidence of Permian (Rotliegendes) sandstones and lavas from Pfalz, Germany; all measurements plotted.

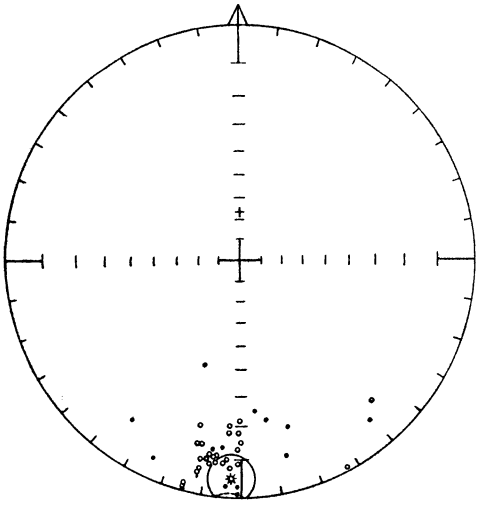


FIG. 12.—The directions of magnetization, mean, and circle of confidence of Permian (Rotliegendes) sandstones and lavas from the Saar; all measurements plotted.

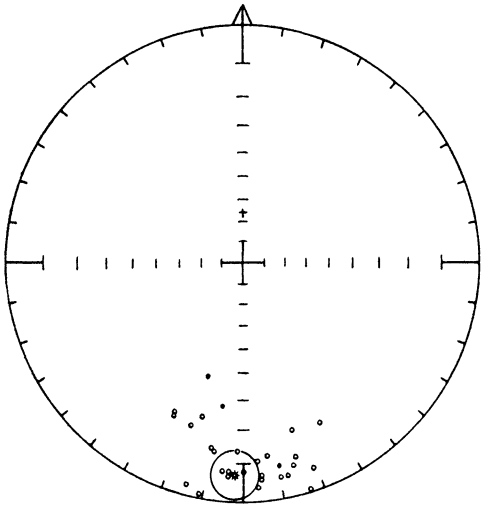


FIG. 14.—The directions of magnetization, mean, and circle of confidence of Permian (Rotliegendes) lavas and sandstones from Baden-Baden, Germany; all measurements plotted.

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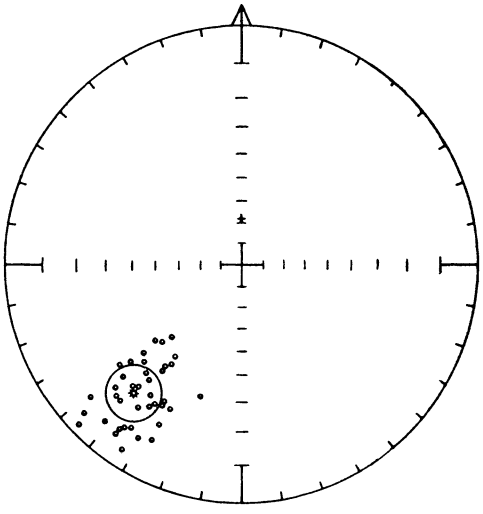


FIG. 15.—The directions of magnetization, mean, and circle of confidence of samples from the Carboniferous Keele beds of Staffordshire; all measurements plotted.

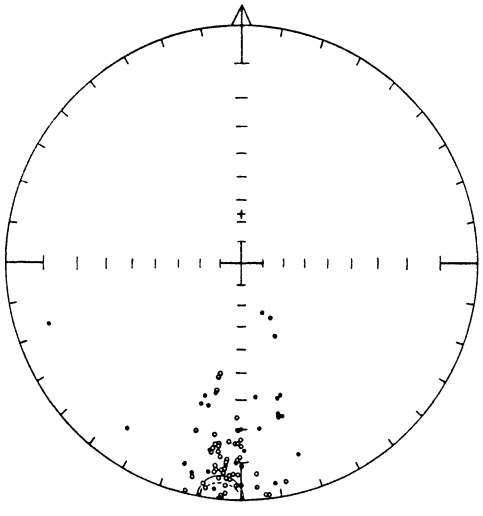


FIG. 17.—The directions of magnetization, mean, and circle of confidence of Stephanian sandstones from the Saar; all measurements plotted.

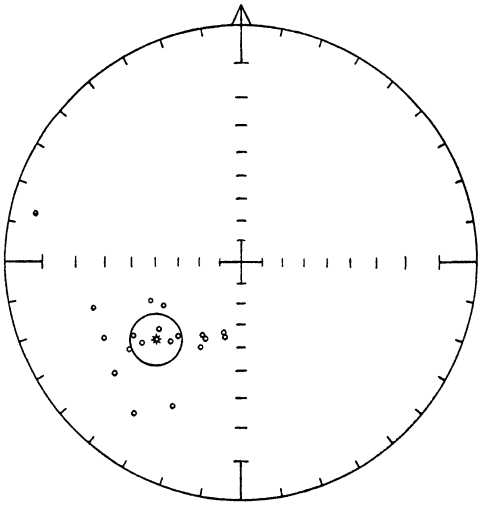


FIG. 16.—The directions of magnetization, mean, and circle of confidence of samples from the Southdean basanite plug, near Jedburgh, southern Scotland; all measurements plotted.

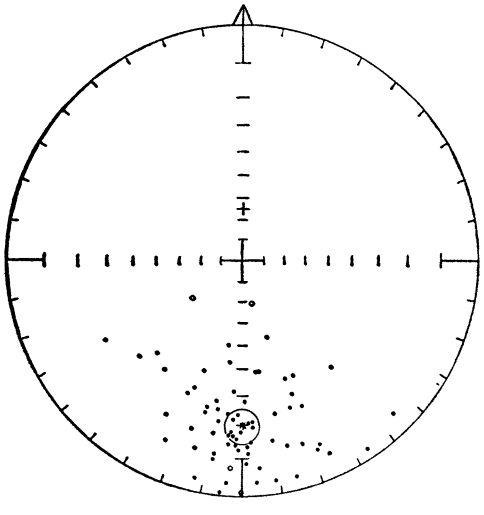


FIG. 18.—The directions of magnetization, mean, and circle of confidence of Stephanian sandstones from Montceaux-les-Mines, France; all measurements plotted.

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have directions of magnetization similar to the Permian. As these collections contain both red and drab beds, the direction is not likely to be due to Permian reddening. It may be suggestive that folding occurred in Great Britain in the time interval between the deposition of the British and European rocks. However, the only European Devonian result which does not differ significantly with the British results seems to militate against this argument.

To this must be added the scattered results obtained by Clegg *et al.* (1957) from groups of igneous rocks varying in age from lavas at horizons well up in the Lower Carboniferous calciferous sandstone of Scotland to Upper Carboniferous intrusives. The results have been assembled in table 5 and are illustrated in figure 20.

7. DEVONIAN—OLD RED SANDSTONE

With the exception of a single group of samples from the Eifel region of Germany (fig. 21), all the paleomagnetic results are from continental Old Red Sandstone deposits of Great Britain. Two new results are included here, one from lower Old Red Sandstone lavas from around Ethie (fig. 22) and the other from sandstones of upper Old Red Sandstone age (fig. 23) collected at Jedburgh in the Scottish borders. The geographical distribution of the sites is shown on figure 24, and the results available are summarized in table 6.

III. DISCUSSION OF RESULTS

The simplest means of depicting paleomagnetic results from a series of rocks is by means of a "polar-wandering curve," in which pole positions from rocks of different ages are linked in chronological order. The alternative method is to plot the area in the latitude and orientation suggested for successive periods (see fig. 27; see also Creer *et al.*, 1958). Sufficiently numerous pole positions are now available for Triassic, Permian, and Devonian periods (fig. 25) to make it worthwhile combining them to give a mean pole position for each period. While there are many problems of method, as a first attempt the pole position from each group of samples has been given unit weight and the vector mean found. These mean pole positions are plotted in figure 26; for other periods the pole positions are those given in Creer *et al.*, 1957. Confidence limits are drawn round all poles except the Jurassic, where with only three results available the confidence limits are very large; this will be clear from inspection of table 2.

No pole position for the Carboniferous has been plotted, hence the break in the curve, because of the problem of the wide scatter in the pole positions (cf. fig. 25, *c*, with fig. 25, *a*, *b*, *d*). The mean incorporating all results given in table 5 is very little different from the Devonian mean, while figure 27, *c*, is based on results from which D_1, D_2, S_1 have been arbitrarily omitted.

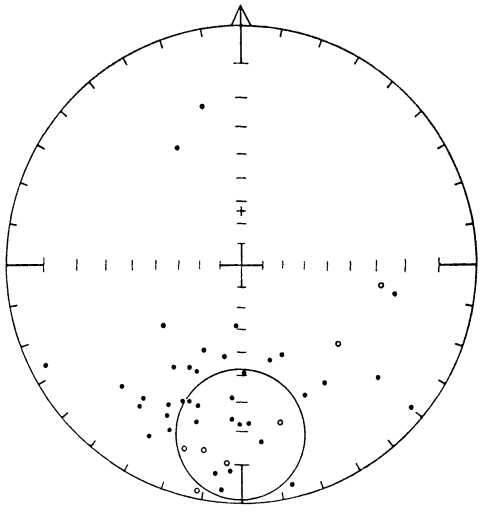


FIG. 19.—The directions of magnetization, mean, and circle of confidence of Stephanian sandstones from St. Étienne, France; all measurements plotted.

One possible reason for the difference between Belshé's original direction and the European results has been suggested, and, until there is more than one pre-Stephanian result from Europe, it will remain a possibility. On the other hand, the dislocation in the apparently smooth curve between upper Old Red Sandstone and lower Per-

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TABLE 5

SUMMARY OF EUROPEAN CARBONIFEROUS PALEOMAGNETIC DATA

	Site			Age and Lithology	Mean Declination (Degrees)	Mean Inclination (Degrees)	α	K	N	S	Ancient Pole	$d\psi$	$d\chi$	Reference
Britain:														
K_1	Kinghorn	56 N.	3 W.....	Lower Carboniferous lavas	200	+38	.....	....	..	...	12 N. 158 E.	....	..	Clegg *et al.*, 1957
K_2	Kinghorn	56 N.	3 W.....	Lower Carboniferous lavas	26	−42	....	....	..	...	8 N. 153 E.	....	..	Clegg *et al.*, 1957
K_3	Kinghorn	56 N.	3 W.....	Lower Carboniferous lavas	20	+15	.....	....	..	..	42 N. 142 E.	....	..	Clegg *et al.*, 1957
J	Jedburgh	55.5 N.	2.5 W.....	Southdean basanite plug	227	−38	9.9	12.6	19	5	41 N. 112 E.	7.4	12	
C	Clee Hills	52.5 N.	2 W.....	Late Carboniferous baked sediments and igneous rocks	200	+15	.....	....	..	...	27 N. 155 E.	....	..	Clegg *et al.*, 1957
T	Tideswelldale	53 N.	1.5 W.....	Upper Carboniferous baked sediments	218	+36	.....	....	..	...	10 N. 142 E.	....	..	Clegg *et al.*, 1957
D_1	Derbyshire	53 N.	1.5 W.....	Lower Carboniferous lavas	48	+47	13.0	....	..	9	46 N. 105 E.	....	..	Belshé 1957
D_2	Derbyshire	53 N.	1.5 W.....	Westphalian sandstones	26	+37	.....	....	103	103	56 N. 136 E.	....	..	Belshé 1957
S_1	Staffordshire	53 N.	2 W.....	Keele beds (red sandstones)	220	−20	9.5	66.6	34	17	37 N. 126 E.	5.2	10	
Germany:														
S_2	Saar	49 N.	7 E.....	Stephanian sandstones	180	0	6	15.8	16	9	41 N. 185 E.	3	6	
France:														
M	Montceaux-les-Mines	47 N.	4.5 E.....	Stephanian sandstones	180	+20	5.3	11.8	66	13	33 N. 185 E.	3	6	
E	St. Étienne	45.4 N.	4.5 E.....	Stephanian sandstones	181	+22	22	3.7	48	10	33 N. 184 E.	12	23	
											Mean pole position = 34 N. 150 E.			

TABLE 6

SUMMARY OF EUROPEAN DEVONIAN PALEOMAGNETIC DATA

	Site			Age and Lithology	Mean Declination (Degrees)	Mean Inclination (Degrees)	α	K	N	S	Ancient Pole	$d\psi$	$d\chi$	Reference
Britain:														
E	Ethie	57 N.	2.5 W.....	Lower Old Red Sandstone lavas	35	+ 5	11.3	24.9	(49)	9	29 N. 136 E.	7	11	
J	Jedburgh	55.5 N.	2.5 W.....	Upper Old Red Sandstone, sandstone	188	+22	8.6	7	46	10	23 N. 169 E.	5	9	
W	S. Wales	52 N.	3 W.....	Lower Old Red Sandstone, sandstone	196	− 4	5	19	..	35*	30 N. 159 E.	3	5	Creer, 1957
P	Portishead	51 N.	4 W.....	Upper Old Red Sandstone, sandstone	208	+10	7.4	28.2	..	12*	29 N. 144 E.	4	8	Creer, 1957
Germany:														
E_1	Eifel	51 N.	6 E.....	Lower-Middle Devonian sandstone	197	−11	23.4	15	(16	4	42 N. 163 E.	4	8	
											Mean pole position = 33 N. 154 E.			

* No. of sites.

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mian times may be a pointer to the "drift" history of Europe.

Many more results are required before it can be ascertained whether significant differences are detectable within a period, implying that the rate of movement, whatever the cause, is sufficiently great.

From a single area it is not possible to decide which form of movement occurred. Upon the discovery that pole positions from rocks of the same age in Europe and North America were significantly different, it became clear that some form of continental drift was inevitable. The measure of the amount of drift is the angular distance between pole positions of like ages, and, by the same token, the period during which separation occurred can be defined. Thus Runcorn (1956*b*) suggested a post-Triassic separation of about 24°, while Du Bois

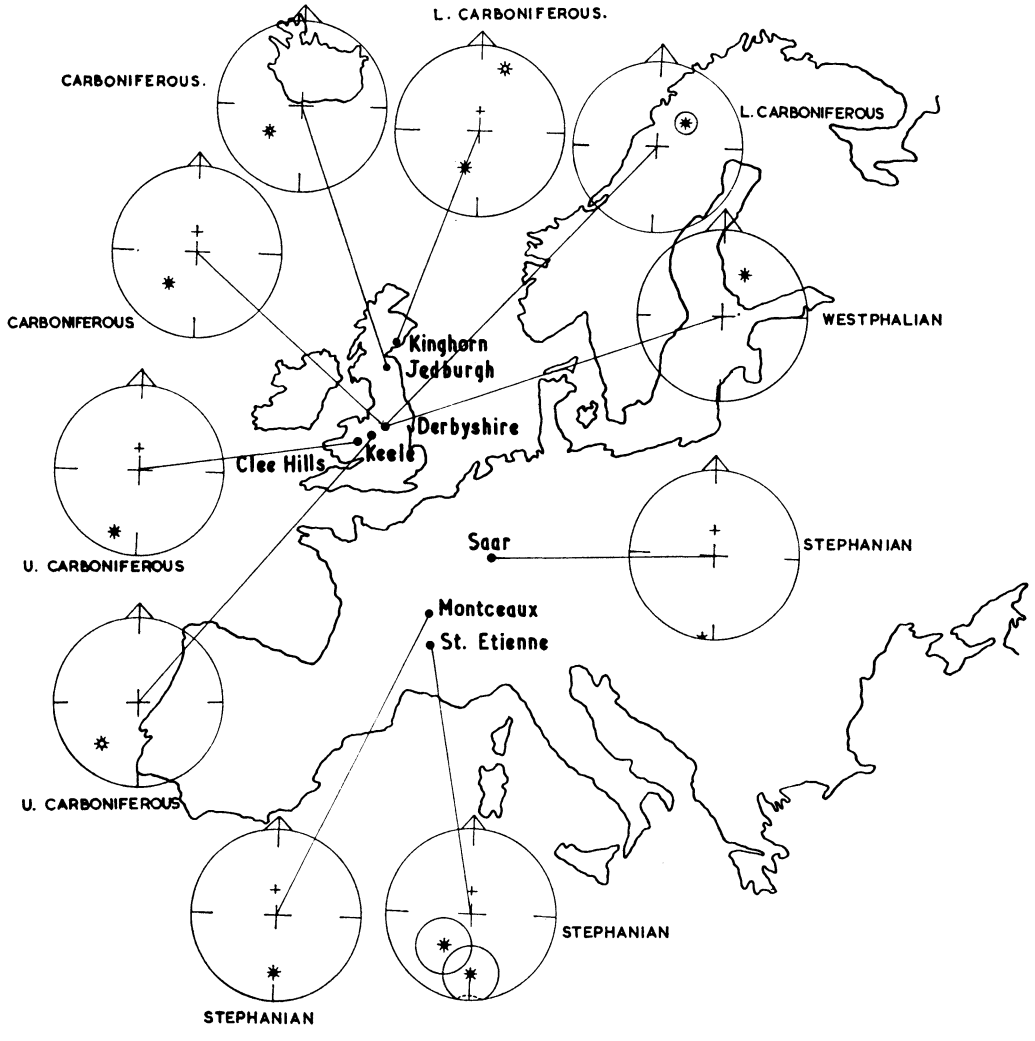


FIG. 20.—Summary of all known Carboniferous paleomagnetic results, indicating the mean magnetic directions for each group of sites. The circle of confidence when known is drawn only where it exceeds 10°. The Second Stephanian mean in the result from St. Étienne was derived from samples collected in the thrust zone, and for this reason it is not included in table 5.

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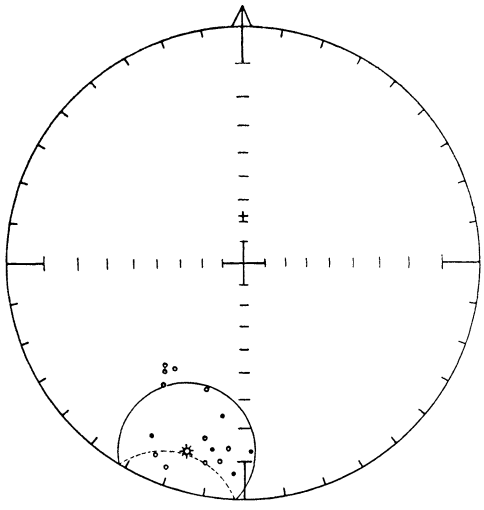


FIG. 21.—The directions of magnetization, mean, and circle of confidence of the lower-middle Devonian red sandstones of the Eifel, Germany; all measurements plotted.

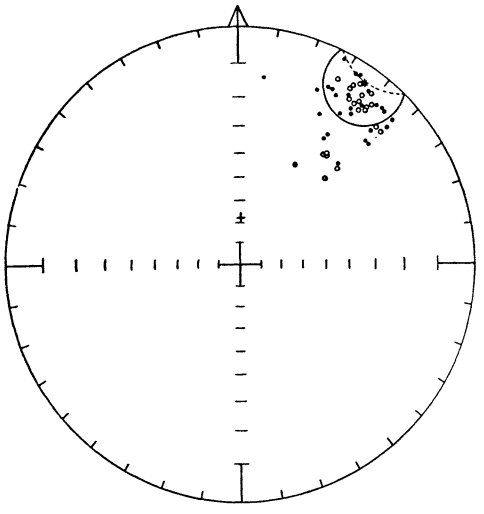


FIG. 22.—The directions of magnetization, mean, and circle of confidence of lower Old Red Sandstone lavas from Ethie, Scotland; all measurements plotted.

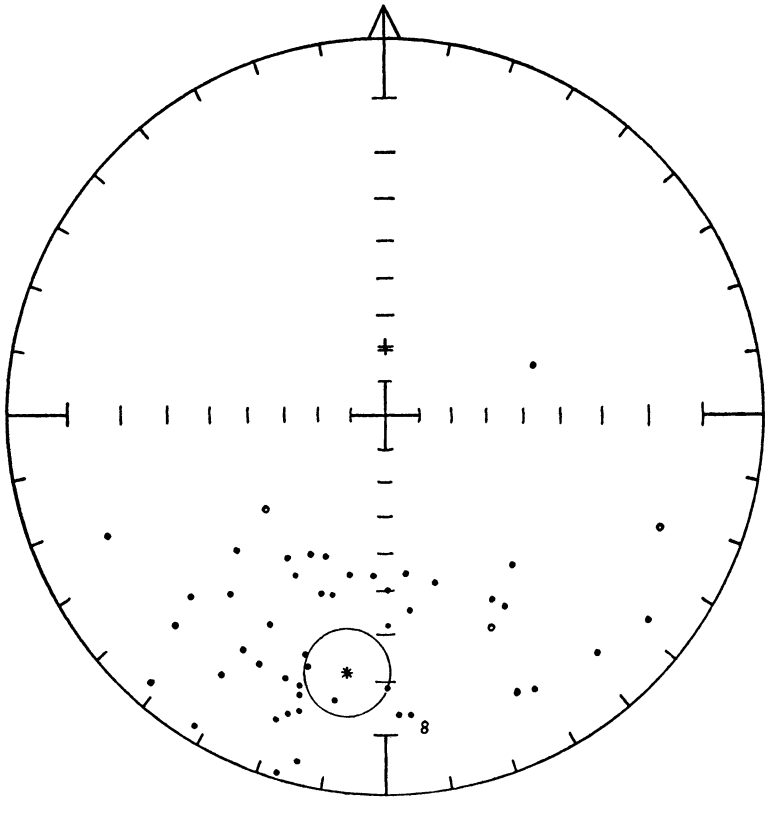


FIG. 23.—The directions of magnetization, mean, and circle of confidence of upper Old Red Sandstone samples from Jedburgh, Scotland; all measurements plotted.

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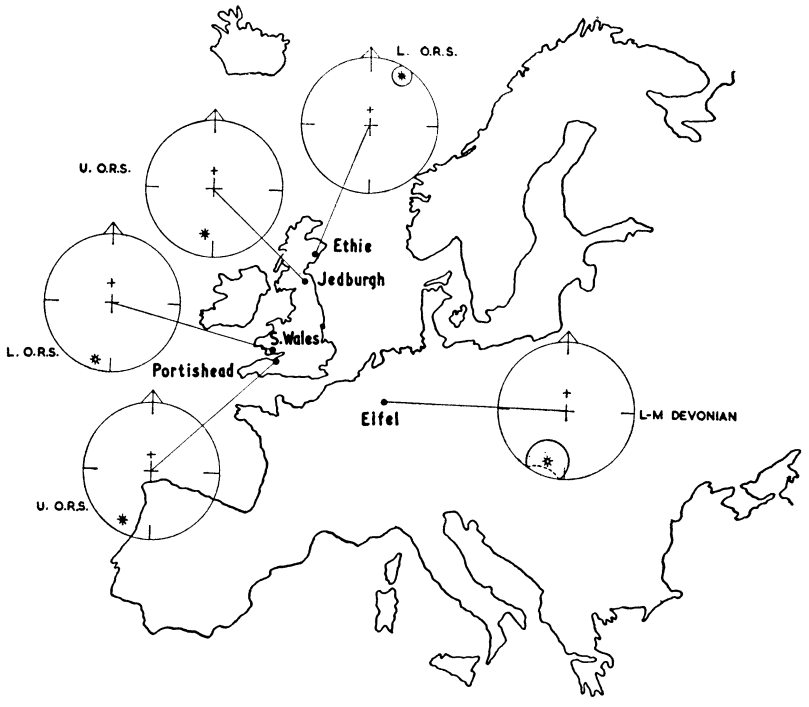


FIG. 24.—Summary of all known Devonian paleomagnetic results, indicating the mean direction for each group of sites. The circle of confidence when known is drawn only where its circle of confidence exceeds 10°.

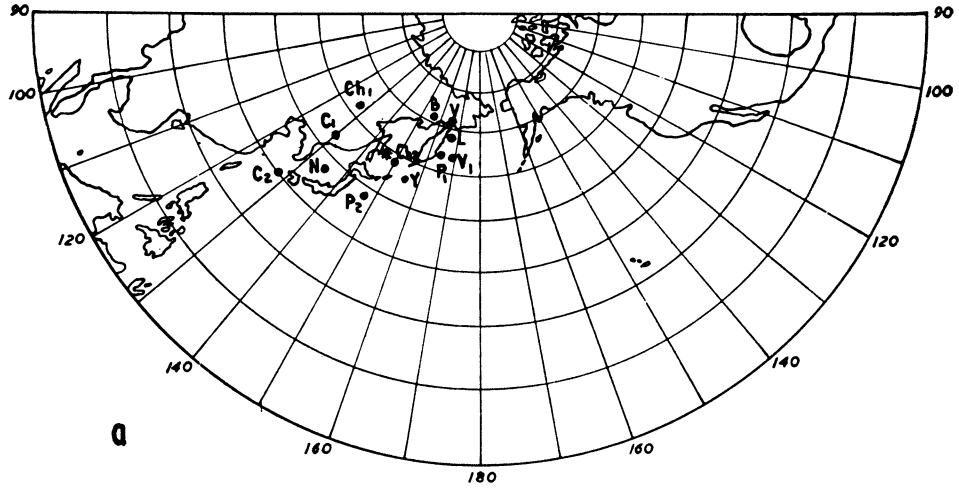


FIG. 25.—Plot of (*a*) Triassic, (*b*) Permian, (*c*) Carboniferous and (*d*) Devonian pole positions based on mean site data for various areas of Europe. Stereographic projection. Symbols correspond to tables 2, 3, 4, and 5.

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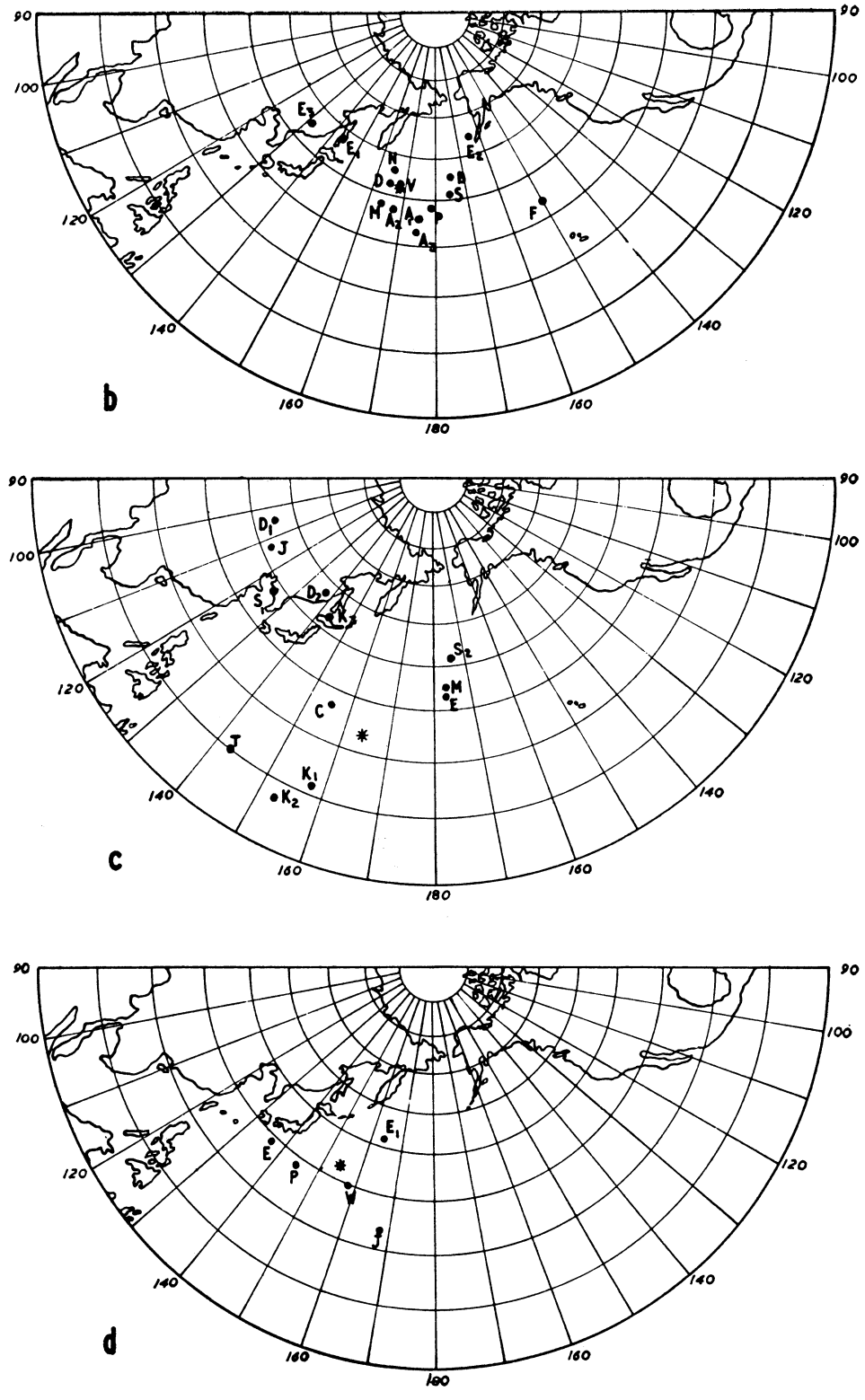


FIG. 25—*Continued*

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(1957) suggested about 40°. The angular distance of 35° between the mean Triassic European pole and the American pole, based on recent results of Collinson and Runcorn (1960), is in good agreement with Runcorn's latest estimate of post-Triassic drift. However, now that the Permian pole position can be regarded as well established both in Europe and in America, their separation of the order of 49° must be regarded as significant and suggestive that between Permian and Triassic times a separation of 14° had already occurred. If the reasons for not quoting a Carboniferous[4] pole position are regarded as valid, there are then no pre-Permian points to check on the earlier drift history.

IV. PALEOCLIMATIC SIGNIFICANCE

Since latitude is an important factor in the climate of any area, paleomagnetism, which makes possible the determination of the ancient latitude and orientation of an area for a given rock series, is directly linked to the study of paleoclimates (see Schove *et al.*, 1958). The results here presented make it clear that the investigated regions of Europe can be treated as a single unit; and figure 27 illustrates Europe in the appropriate position for Triassic, Permian, and Devonian times. As already discussed, the Carboniferous position is arbitrary, and, if all results were used, the result would differ little from the Devonian.

Plotted on figure 27 is the information, derived from geology, which is commonly regarded as being climatically significant (Opdyke, personal communication). It will be seen that there is general agreement between the paleomagnetic latitude and the climatic evidence because evaporites, red beds, coals, reef corals, and bauxite deposits are all evidence of hot climates. This conclusion is strengthened by the study (from dune-bedded sandstones) of ancient wind directions. Opdyke and Runcorn (1960) show that these winds when related to paleomagnetic reconstructions are consistent in latitude and direction with the Trade Wind belt of the Northern Hemisphere.

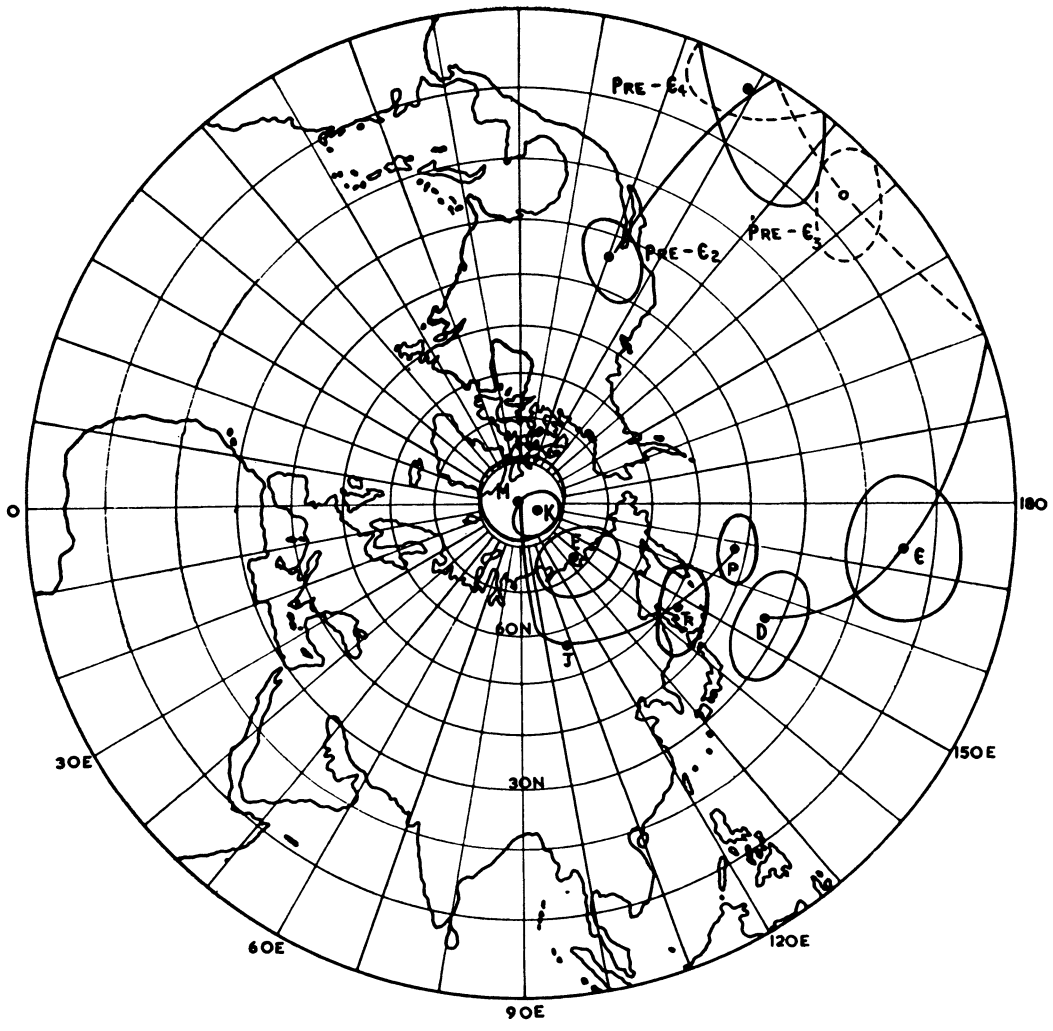
[4] The separation between the mean European and American (Naco sandstone, Creer *et al.*, 1957) Carboniferous poles is 24°.

V. CONCLUSIONS

It has been established that for certain periods the directions of magnetization form a consistent pattern; and for the Triassic, Permian, and Devonian periods, mean pole positions have been calculated. The predicted latitudes and orientations in all three periods are in agreement with the known paleoclimatic data and with what is known of wind directions in Triassic and Permian times. Miocene and later results are coincident with the dipole-field assumption. Satisfactory mean pole positions for Cretaceous and Jurassic rocks are handicapped by the paucity of results, while the scattered Carboniferous pole positions make interpretation difficult without further study. It is noticeable that the apparent "dislocation" of the curve linking pole positions is Carboniferous in age, while the separation of North America and Europe would appear to date from Permian times.

ACKNOWLEDGMENTS.—For much help and friendly advice from his colleagues in Newcastle, and especially to S. K. Runcorn, the author expresses his gratitude. Thanks are also due to E. Irving and N. D. Opdyke of the Australian National University, Canberra, for their help. The field collections could not have been made without the help of Gogeul, Minoux, de Maistre, Feys, Lanly, and Bolfa in France and of Kürsten, Güthorl, Bilharz, Sauer, Westphal, Bank, and Prange in Germany.

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LEGEND

E	EOCENE.	€	CAMBRIAN.
K	CRETACEOUS.	PRE - €4	PRE CAMBRIAN LONGMYNDIAN.
J	JURASSIC.	PRE - €3	PRE CAMBRIAN UPPER TORRIDONIAN.
Tr	TRIASSIC.	PRE - €2	PRE CAMBRIAN LOWER TORRIDONIAN.
P	PERMIAN.		
D	DEVONIAN.		

FIG. 26.—Polar stereographic projection showing mean pole positions, in chronological order, of European rocks.

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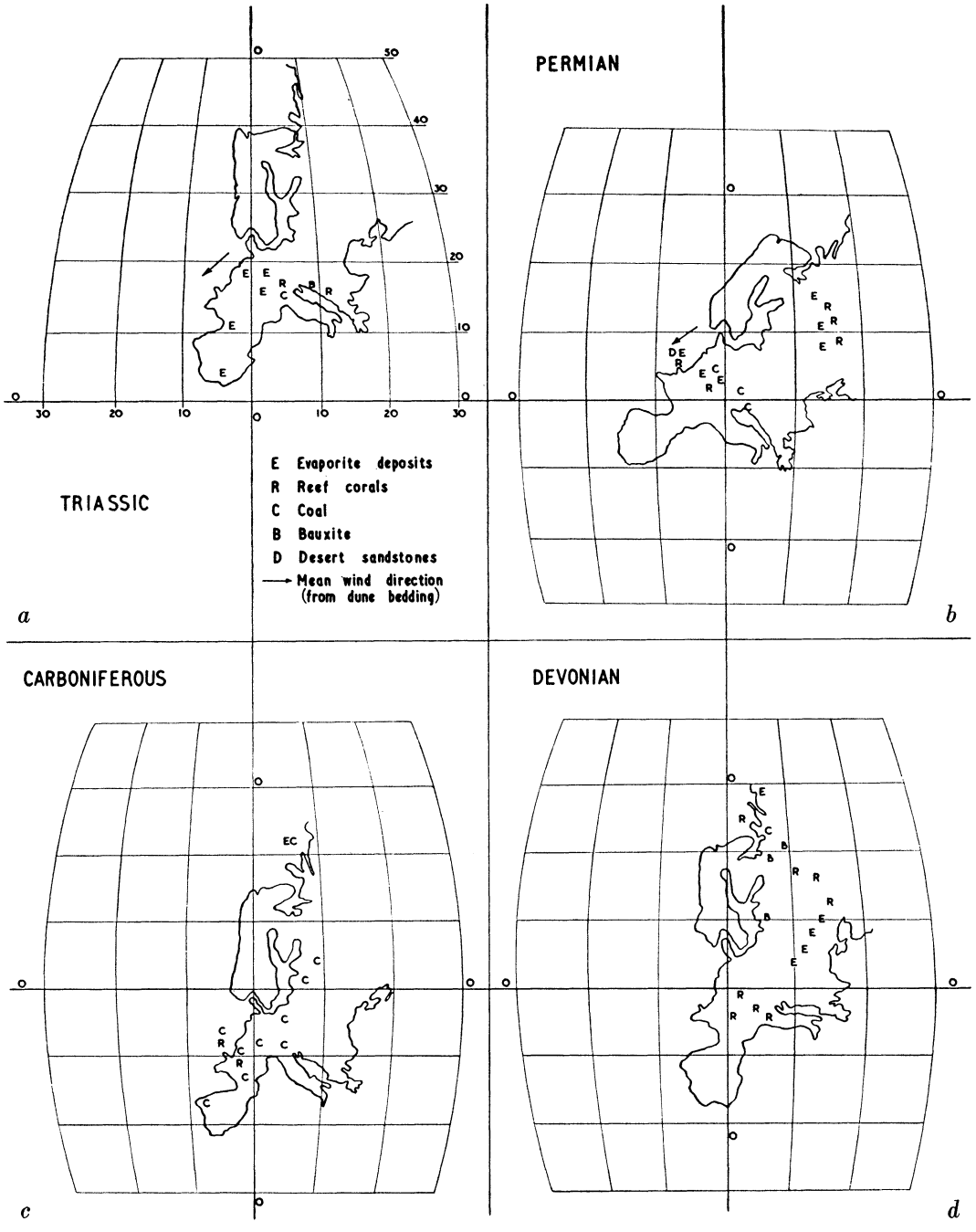


FIG. 27.—The restoration of Europe to the latitudes and orientations suggested by Triassic, Permian, Carboniferous, and Devonian paleomagnetic data incorporating indications of climate derived from geology.

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APPENDIX

The new results given in this paper were derived from samples collected in the following localities:

Period	Type of Outcrop	Locality
Tertiary:		
Vogelsberg	Quarries	Breitenborn, Grebenhaim, Nieder Ofleiden, Oberwiddersheim, Gläshutten Schotten, Eschenrod, Windelhausen, Lissberg, Alsfeld, Lauterbach, Weilberg
Bonn	Quarries	Walsdorf, Huhnerberg, Rottbitze, Niederbachem, Berkum, Arzdorf, Nieder Dollendorf, Montabaur, Muffendorf, Königswinter
Triassic:		
Lörrach (L)	Quarries	Fahrnau wood, quarry near station and one near fork in path through woods
Vogelsberg (V)	Quarries	Hauswaur, Mernes, Gelnhausen
Pfalz (P_1)	Quarries	Kaiserslautern (N.E. outskirts), Hauptstuhl
Pfalz (P_2)	Road cuttings, quarry	Rieschweilor (N. and E.), Zweibrucken, Freiburg (Lorettohof)
Baden-Baden-Freiburg (B)	Natural outcrop	Schromberg (path to Schlossberg), road between Hündoeck to Unterstamatt
Vosges	Road cuttings, quarries	Col. du Haut St. Jacques, Liesbach, Épinal (Urmenil and carrière de la Vierge)
Permian:		
Saar (S)	Quarries, natural outcrops	St. Wendel, Oberlinxweiler, road between Grönig and Selbach, northern outskirts of Tholey
Pfalz (P)	Quarries	Idar-Oberstein (Setz, Völlnersbach), Nohfelden, Ellweiler, roadside quarry between Diensweiler and Hoppstedten, Martinstein, roadside quarry near Traison, near Fishbach, Neiderbrombach
Baden-Baden (B)	Road cuttings, quarries	Near Sulzbach, and near Ebersteinburg, Leissberg, Seelechstrasse, Ruhstein, Edelfrauengrabe in vicinity of Baden-Baden
Freiburg (F)	Quarries	Denzlingen, between Sexau and Lörsch, Burgholz between Scheuermatt and Schweigmatt
Carboniferous:		
Staffordshire (S_1)	Quarries, road cuttings	Near Newcastle (Staffordshire) Clockhouse Drive, Two-mile lane, Honeywell Farm
Saar (S_2)	Quarries	Ottweiler, Dorrenbach, Illingen, between Wiesbach and Urchtilfangen, Wellesweilor, Fürth
Montceaux-les-Hines (M)	Quarries	St. Hélène, St. François
St. Étienne (S_3)	Road cuttings, quarries	Talaudière, puit la Croix, road near La Chana, Ricolin, near St. Charmont and La Pacotière near rifle range
St. Étienne	Road cuttings, quarries (thrust zone)	Carrière de la Poule Noire, Mont Rombert
Devonian:		
Ethie (E)	Coast outcrop	Ethie Haven, and south of Ferryden lighthouse
Jedburgh (J)	Natural outcrop	Various points in river Jed south of Jedburgh
Eifel (E_1)	Quarry	Northwest of Kirchheim near Euskirchen

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